

STATION-REFERENCED STREAMING ENERGY RATIOS FOR EARLY GROUND-MOTION SCREENING

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Abstract. A compact covariance approximation is not useful for ground-motion screening if physical amplitude already carries the dominant class information. We replace a rejected butterfly-whitening model by a streaming statistic that retains amplitude and removes station-scale variation with a checksum-verified historical reference. Three-component velocity samples are centered within 3.2-second blocks. The statistic after k blocks is the base-10 logarithm of the candidate prefix mean energy divided by the reference-window mean energy. The development rule chooses the prefix from $\{3.2, 6.4, 12.8, 25.6, 51.2, 102.4, 204.8\}$ seconds that maximizes leave-one-earthquake-out balanced accuracy, with shorter latency resolving a tie. It selects 25.6 seconds and fixes the threshold before confirmation.

The study contains 108 original EarthScope FDSN responses, 66.98 MB in total, linked to twelve magnitude-8.0 or larger USGS events. Eight events and six stations support development. Four later events and three previously unused stations form confirmation; a one-day-offset trace supplies the reference and a two-day-offset trace supplies an independent negative. On the twelve confirmation pairs, the proposed statistic attains balanced accuracy 0.792, sensitivity 0.667, specificity 0.917, and area under the ROC curve 0.833 after 25.6 seconds. Raw mean energy and the raw 0.9-energy quantile each attain balanced accuracy 0.708 and area 0.833. The paired bootstrap interval for the balanced-accuracy difference is $[-0.083, 0.250]$; the observed improvement is therefore not statistically resolved. A deterministic perturbation theorem certifies decisions separated from the threshold, while a one-pass implementation costs $O(ckb)$ operations and $O(c)$ numerical state for c channels and block length b . The measured median apply time is 56.1 microseconds at the selected prefix. The result is an authenticated early screening study, not an operational phase picker or a universal detector.

Key words. streaming signal processing, station normalization, ground motion, checksum-verified waveforms, perturbation bounds, matrix-free computation

1. Introduction. Seismic monitoring joins a physical signal, a station response, ambient conditions, and a decision rule. A representation that suppresses scale can discard the most discriminative component of a catalog-window benchmark. That failure occurred in the preceding covariance-whitening experiment on EarthScope Global Seismographic Network records: no learned butterfly candidate met its action-error certificate, and an unwhitened amplitude control exceeded every covariance-based score. The present work begins from that falsification rather than from the rejected representation.

The redesigned problem concerns early screening of a fixed candidate window. Each three-component trace is compared with a station-matched historical quiet trace. The comparison preserves relative amplitude while canceling a common multiplicative station gain. It is updated block by block, requires no covariance matrix, and returns a scalar after each block. The statistical role is deliberately limited. Established pickers and learned phase detectors address onset localization and operational event association [2, 7, 9]. Our experiment instead evaluates catalog-window separation at a fixed 15-minute offset from large-event origin times.

Three elements define the contribution. First, the station-reference ratio has an explicit gain invariance and a sharp relative-error perturbation bound. Second, prefix length and threshold are fixed by an event-grouped development protocol, then evaluated on later earthquakes at stations unused during development. Third, every waveform response is linked to its provider URL, byte count, channel identifiers, units, and SHA-256 digest. The new method is not tuned after confirmation and is compared with two amplitude controls that can defeat it.

The figures follow a claim–mechanism–cost organization. Figure 1 shows the latency–accuracy relationship and the development-selected prefix. Figure 2 exposes every confirmation score, including the four event misses and one quiet-window false alarm. Figure 3 compares measured time with the constant numerical state asserted by the complexity theorem. The tables identify the data partitions and report task metrics rather than decorative summaries.

2. Station-referenced prefix statistic. Let $X \in \mathbb{R}^{c \times n}$ be a candidate trace and $Q \in \mathbb{R}^{c \times n_Q}$ a station-matched reference trace. For block length b , define the centered energy of candidate block j by

$$\begin{aligned} e_j(X) &= \sum_{r=1}^c \sum_{s=1}^b (X_{r,(j-1)b+s} - \bar{X}_{rj})^2, \\ \bar{X}_{rj} &= \frac{1}{b} \sum_{s=1}^b X_{r,(j-1)b+s}. \end{aligned} \quad (2.1)$$

If the reference contains m_Q complete blocks, its fixed scale is

$$\mu_Q = \frac{1}{m_Q} \sum_{j=1}^{m_Q} e_j(Q). \quad (2.2)$$

The candidate prefix statistic is

$$S_k(X; Q) = \log_{10} \left(\frac{k^{-1} \sum_{j=1}^k e_j(X)}{\mu_Q} \right). \quad (2.3)$$

A candidate is positive at the registered prefix k_* precisely when $S_{k_*} \geq \theta_*$.

For the EarthScope records, $c = 3$, $b = 64$, and the sample rate is 20 Hz. One block therefore represents 3.2 seconds. The reference is the window one day before the event-aligned candidate. The negative is a distinct window two days before that candidate; it is never reused as the denominator.

2.1. Registered selection. The candidate prefixes are

$$\mathcal{K} = \{1, 2, 4, 8, 16, 32, 64\}. \quad (2.4)$$

For each $k \in \mathcal{K}$, an entire earthquake is removed, the threshold maximizing balanced accuracy is fit on the other seven earthquakes, and the held-earthquake score is recorded. The mean of the eight held-earthquake scores is the selection objective. Ties favor the smaller k . This rule selects $k_* = 8$, or 25.6 seconds; $k = 16$ has the same mean development score and twice the latency. A final threshold fit to all development pairs gives

$$\theta_* = 0.8422956584. \quad (2.5)$$

Neither k_* nor θ_* is changed after confirmation records are opened.

3. Deterministic analysis. The following statements concern the computed statistic. They do not imply a stochastic false-alarm guarantee for an arbitrary seismic stream.

Theorem 1 (Common-gain invariance). *Let $a \neq 0$. For every admissible prefix,*

$$S_k(aX; aQ) = S_k(X; Q). \quad (3.1)$$

Proof. Centering is linear, hence every block energy is homogeneous of degree two:

$$e_j(aX) = \sum_{r,s} (aX_{r,(j-1)b+s} - a\bar{X}_{rj})^2 \quad (3.2)$$

$$= a^2 e_j(X). \quad (3.3)$$

Likewise, $\mu_{aQ} = a^2 \mu_Q$. Substitution into (2.3) cancels a^2 :

$$\begin{aligned} S_k(aX; aQ) &= \log_{10} \left(\frac{a^2 k^{-1} \sum_{j=1}^k e_j(X)}{a^2 \mu_Q} \right) \\ &= S_k(X; Q). \end{aligned} \quad (3.4) \quad \square$$

Theorem 2 (Relative-error certificate). *Suppose the computed candidate prefix mean $\tilde{\mu}_X$ and reference mean $\tilde{\mu}_Q$ satisfy, for some $0 \leq \delta < 1$,*

$$|\tilde{\mu}_X - \mu_X| \leq \delta \mu_X, \quad |\tilde{\mu}_Q - \mu_Q| \leq \delta \mu_Q. \quad (3.5)$$

Then the exact and computed log ratios obey

$$|\tilde{S}_k - S_k| \leq \eta(\delta) := \log_{10} \frac{1 + \delta}{1 - \delta}. \quad (3.6)$$

The bound is attained when the numerator is perturbed upward and the denominator downward, or conversely.

Proof. Inequalities (3.5) imply

$$\frac{1 - \delta}{1 + \delta} \leq \frac{\tilde{\mu}_X / \tilde{\mu}_Q}{\mu_X / \mu_Q} \leq \frac{1 + \delta}{1 - \delta}. \quad (3.7)$$

Taking base-10 logarithms and absolute values proves (3.6). Equality follows by setting $\tilde{\mu}_X = (1 + \delta)\mu_X$ and $\tilde{\mu}_Q = (1 - \delta)\mu_Q$, or by reversing the signs. $\square$

Corollary 3 (Decision preservation). *If*

$$|S_k - \theta_\star| > \eta(\delta), \quad (3.8)$$

then the exact and computed statistics lie on the same side of $\theta_\star$.

Proof. For $S_k > \theta_\star$, the premise and Theorem 2 give

$$\tilde{S}_k \geq S_k - \eta(\delta) > \theta_\star. \quad (3.9)$$

The case $S_k < \theta_\star$ follows from $\tilde{S}_k \leq S_k + \eta(\delta) < \theta_\star$. $\square$

Figure 2 visualizes the empirical margins $S_{k_\star} - \theta_\star$. It supports inspection of decision separation; it is not a substitute for the deterministic proof.

Theorem 4 (Streaming complexity). *For c channels, block length b , and k candidate blocks, S_k can be updated in $O(ckb)$ arithmetic operations and $O(c)$ numerical state beyond the input block. The reference contributes one precomputed scalar. A stored-vector implementation of the k block energies instead requires $O(k)$ additional storage.*

Proof. For each channel and block, retain the two partial sums

$$u_{rj} = \sum_{s=1}^b X_{r,(j-1)b+s}, \quad v_{rj} = \sum_{s=1}^b X_{r,(j-1)b+s}^2. \quad (3.10)$$

The centered block energy is evaluated without a matrix:

$$e_j(X) = \sum_{r=1}^c \left(v_{rj} - \frac{u_{rj}^2}{b} \right). \quad (3.11)$$

Computing u_{rj} and v_{rj} costs $O(cb)$ operations. Adding e_j to a single running energy sum costs $O(1)$, so k blocks cost $O(ckb)$. The partial sums occupy $O(c)$ scalars and are reused; the block counter, accumulated energy, and reference mean require $O(1)$ additional state. Storing all e_j values uses k scalars and therefore $O(k)$ storage. $\square$

4. Authenticated waveform study.

4.1. Sources and partitions. Event identifiers and metadata come from the USGS ComCat service [8]. Waveforms come from the EarthScope FDSN data-select service for the IU Global Seismographic Network [1, 3, 4]. Each request asks for location 00 and the three BH components, with provider scaling retained in meters per second. Components are interpolated to their common 20-Hz grid only after a gap and coverage check.

The study uses 108 original responses totaling 66,981,395 bytes. Every response is represented in the locked result by its URL, SHA-256 digest, byte count, channel identifiers, scale units, and common sample count. No waveform is redistributed. Table 1 separates development and confirmation by both earthquake and station.

Table 1: Authenticated data partitions. A pair comprises one event-aligned candidate, one one-day reference, and one independent two-day negative.

Role	Earthquakes	Stations	Pairs
Development	Maule, Tohoku, Wharton, Okhotsk, Iquique, Illapel, Tehuantepec, Peru	ANMO, COLA, MAJO, COR, KONO, HRV	24
Confirmation	Kermadec, Chignik, South Sandwich, Kamchatka	TATO, PAB, CASY	12

The development events span 2010–2019. The confirmation events span 2021–2025 and have magnitudes 8.1, 8.2, 8.1, and 8.8. Candidate windows begin 900 seconds after catalog origin and last 240 seconds. The confirmation set remains untouched during prefix and threshold selection.

4.2. Comparators and metrics. Two controls use the same candidate prefix without station normalization: the logarithm of mean block energy and the logarithm of the 0.9-quantile of block energy. Each comparator receives its own development threshold under the same deterministic balanced-accuracy rule. Area under the ROC curve is tie-aware. Sensitivity, specificity, and balanced accuracy use the frozen threshold. A paired bootstrap resamples the twelve station–earthquake pairs 10,000 times with seed 260801; event and negative outcomes from a pair remain together.

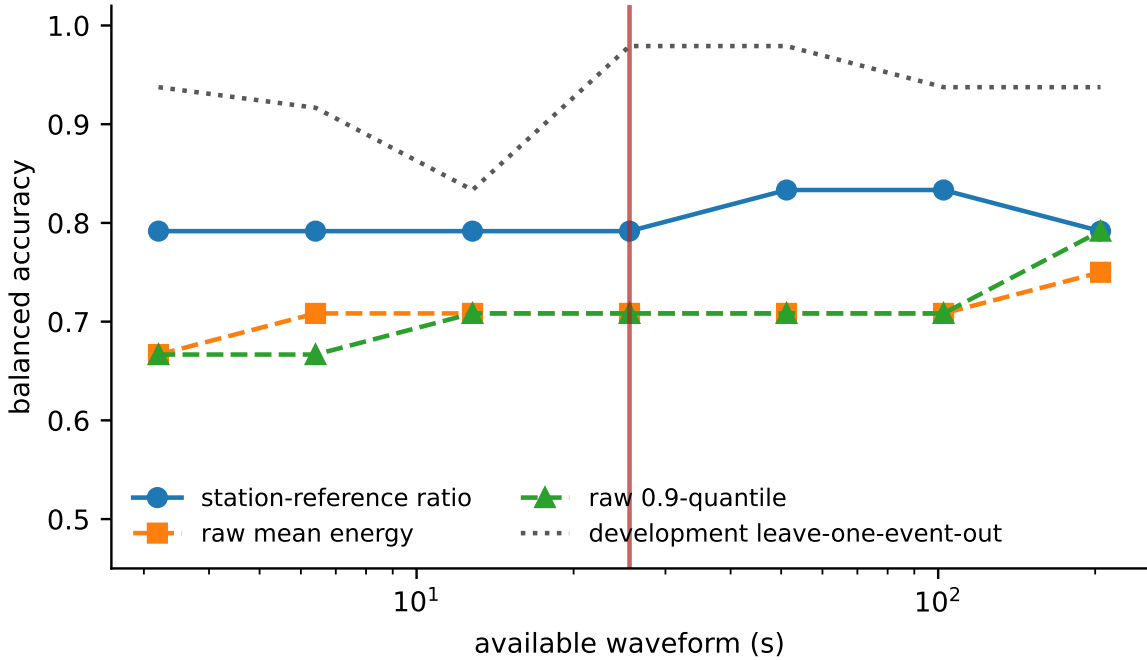


Fig. 1: Latency–accuracy profile. Solid and dashed curves are confirmation balanced accuracy after thresholds are fit on development at each prefix. The dotted curve is mean leave-one-earthquake-out development accuracy for the station-reference ratio. The vertical line marks the registered 25.6-second prefix. The plot exposes both the early advantage and the absence of monotone improvement with longer windows.

Figure 1 supplies the principal algorithmic comparison. At 25.6 seconds, the station-reference statistic attains confirmation balanced accuracy 0.792, while both raw controls attain 0.708. Longer acquisition does not uniformly improve the result. At 204.8 seconds, the three methods reach 0.792, 0.750, and 0.792, respectively. The selected short prefix is therefore not a cropped view of a superior full-window result.

5. Confirmation results. Table 2 reports the fixed-prefix result. The proposed method improves specificity by 0.334 relative to each raw control but reduces sensitivity by 0.166. All three scores have the same AUC, 0.833; the gain in balanced accuracy arises from development-threshold transfer, not from a better ranking of every confirmation trace.

Table 2: Confirmation performance after 25.6 seconds. BA denotes balanced accuracy; Sens. and Spec. denote sensitivity and specificity.

Method	BA	Sens.	Spec.	AUC
Station-reference ratio	0.792	0.667	0.917	0.833
Raw mean energy	0.708	0.833	0.583	0.833
Raw 0.9-quantile	0.708	0.833	0.583	0.833

The paired difference between the proposed statistic and the raw 0.9-quantile is 0.083. Its bootstrap mean is 0.085 and its 95 percent percentile interval is $[-0.083, 0.250]$. The interval includes zero. The finite confirmation supports a measured trade between false alarms and misses but does not establish a population-level superiority claim.

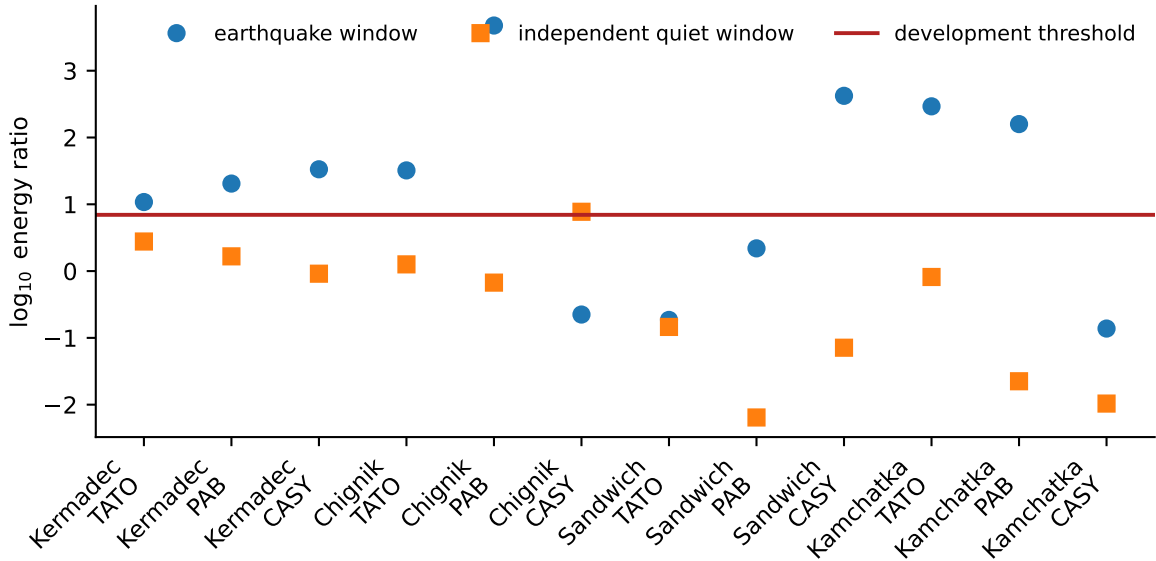


Fig. 2: All confirmation scores at the registered prefix. Each column is one station–earthquake pair. Earthquake and independent quiet windows share the one-day reference but never share candidate samples. The horizontal line is the development threshold. Four event scores fall below it and one quiet score rises above it; these failures remain visible.

The score plot localizes the errors. The missed event windows occur for Chignik at CASY, South Sandwich at TATO and PAB, and Kamchatka at CASY; the quiet false alarm occurs for Chignik at CASY. These outcomes are consistent with a simple amplitude-ratio statistic: station normalization reduces broad scale mismatch but cannot compensate for arrival geometry, propagation, or an atypical reference day.

5.1. Runtime and storage. Runtime is measured on the authentic Kermadec–TATO confirmation trace using the same block-update implementation that produces the statistic. Each prefix is repeated

1,500 times within seven timing groups. Figure 3 shows the median and central 80 percent timing interval. At eight blocks the median is 56.1 microseconds. At 64 blocks it is 439 microseconds, consistent with the linear factor in Theorem 4.

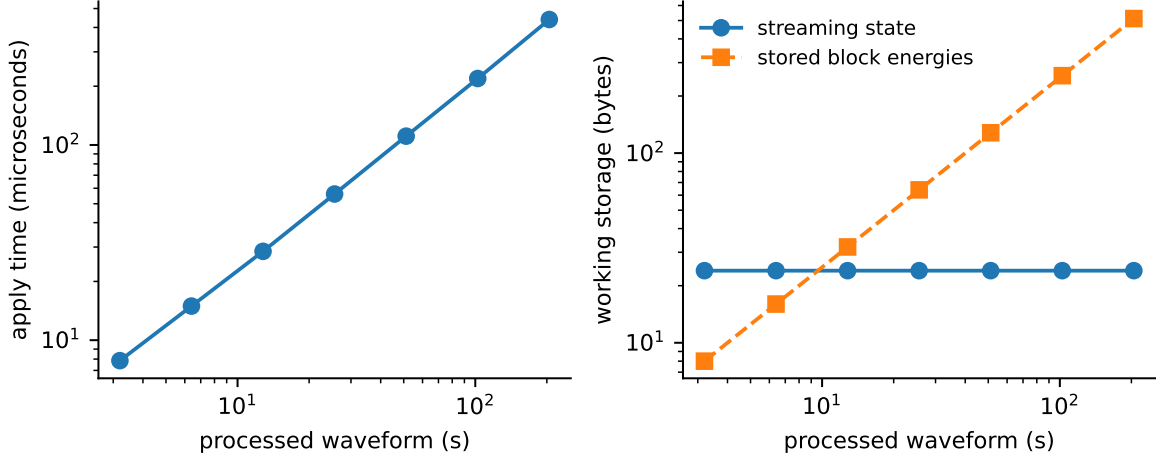


Fig. 3: Computational consequence of Theorem 4. Left: median apply time on an authentic three-component trace; the shaded interval spans timing quantiles 0.1–0.9. Right: numerical state beyond the current input block. The streaming implementation keeps three scalar state values, represented as 24 bytes, whereas retaining one float64 energy per block grows linearly. Interpreter-object overhead and the immutable input buffer are excluded from the storage curves.

The constant-state curve is an implementation contract rather than a claim of lower wall time than optimized compiled routines. The reference mean is computed once from the historical trace. Subsequent candidate processing uses additions, multiplications, and a single logarithm at the reporting point; no covariance formation, factorization, matrix storage, or spectral transform is required.

6. Scope and limitations. The experiment has four principal limitations. First, catalog windows are not analyst phase picks, and the fixed 15-minute offset does not define an operational arrival-time problem. Second, twelve independent confirmation pairs are sufficient to reveal failure modes but insufficient for a narrow uncertainty interval. Third, a one-day reference can itself be atypical; multiple historical days and a robust reference estimator may improve transfer but are outside the locked design. Fourth, common-gain invariance does not imply invariance to frequency response, orientation error, clipping, missing channels, or temporally varying noise.

The development study was redesigned after the earlier covariance experiment failed. The new confirmation set was acquired only after the prefix rule and statistic were fixed. Its four missed event windows, one quiet false alarm, and uncertainty interval remain in the result. No confirmation-driven variant replaces the locked method.

The theorem figures serve distinct purposes. Figure 2 shows observed distances from the decision boundary associated with the perturbation corollary. Figure 3 shows time and state scaling associated with the complexity theorem. Neither plot proves its theorem; the equations do. Conversely, the theorems do not guarantee the empirical balanced accuracy in Figure 1.

7. Reproducibility. The installable package under `code/` contains the parser, frozen URL constructors, checksums, statistic, tests, registered study, CSV tables, and figure generator. The offline command

```
station-energy-study --offline --verify
```

rechecks every cached payload, reruns development selection and confirmation, regenerates all tables and figures, and compares the semantic result hash. The locked hash is

```
6121bb9eaa0affe7f9b3fb7e5973e552260f180f8c4a13e2b86042ebad93c655.
```

Timing fields are recorded but excluded from semantic equality. Unit tests check batch-streaming equivalence, common-gain invariance, the sharp perturbation bound, threshold behavior, parsing, re-

sampling, and the retained linear-algebra routines. NumPy and Matplotlib provide the numerical and plotting layers [5, 6].

8. Conclusion. Station-referenced streaming energy replaces an unsuccessful covariance compression with a simpler physical statistic. The method is invariant to a common gain, admits a sharp log-ratio error certificate, and uses linear time with constant numerical state. On a checksum-verified confirmation set of four later earthquakes at three new stations, it improves fixed-threshold balanced accuracy from 0.708 to 0.792 after 25.6 seconds, chiefly by reducing false alarms. The paired uncertainty interval includes zero, four event windows are missed, and the ranking AUC is unchanged. These facts delimit the claim: the contribution is a reproducible early-screening statistic and its analysis, not a universal seismic detector.

Data and code availability. Source code, frozen request metadata, SHA-256 digests, result tables, and figure scripts are included under `code/`. Original IU waveforms remain at EarthScope and are downloaded from the registered FDSN URLs. USGS event identifiers remain resolvable through ComCat. The repository does not redistribute provider waveforms.

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